

DAY-5

Software Development Life Cycle (SDLC)

- **Software Development Life Cycle (SDLC)**

Software Development Life Cycle (SDLC) is the process followed to develop software from collecting requirements to maintaining the application after release.

- **Phases of SDLC**

1. Requirement Analysis
2. System / Functional Design
3. Development Phase
4. Testing Phase
5. Deployment Phase
6. Maintenance Phase

1. Requirement Analysis

This is the first phase of SDLC.

- The client explains all the project requirements.
- The Business Analyst (BA) gathers all the requirements.
- These requirements are documented in a document called **Functional Specification Document (FSD)**.
- The FSD contains the complete functional requirements collected from the client.
- This document is shared with the development team.

Functional Specification Document (FSD)

- Contains the entire requirement from the client.
 - Explains what the system should do.
 - Used by developers, testers, and other teams.
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2. System / Functional Design

After understanding the requirements:

- Developers create another document.
 - This document contains:
 - Technologies to be used
 - System architecture
 - Database design
 - Application flow
 - Design details
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3. Development Phase

- Developers write the application code.
 - Development is first done in the **Local Server**.
 - Initial testing is also done in the local server.
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4. Testing

Developers perform basic testing before giving the application to testers.

Testers

- Prepare **Test Cases**.
 - Test the application.
 - Find bugs and report them to developers.
 - After fixing all bugs, the application is ready for deployment.
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5. Deployment Phase

The application is moved through different servers.

Three Main Servers

1. Development Server

- Developers develop the application.

- Initial testing is done here.

2. Quality (QA) Server

- Testers perform complete testing.
- Quality Assurance team verifies the application.

3. Production Server

- Final server.
 - End users can use the application.
 - Live environment.
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6.Maintenance Phase

After deployment:

- Fix bugs found by users.
 - Add new features.
 - Improve performance.
 - Update the application whenever required.
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• Company Teams

HR Team

Responsibilities:

- Recruitment
 - Salary discussion
 - Employee-related activities
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IT Team

Responsibilities:

- Database management
- Technical support
- Server maintenance

- Application management
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Admin and Management

Responsibilities:

- Internal company operations
 - Salary management
 - Company-oriented administrative work
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Product Manager vs Project Manager

Product Manager

Defines product vision

Creates product roadmap

Decides product goals

Focuses on customer needs

Project Manager

Manages project execution

Manages project timeline

Allocates resources

Ensures timely project delivery

- **Software Development Models**

1. Waterfall Model

Waterfall is a **Sequential Model**.

Each phase starts only after the previous phase is completed.

Advantages

- Easy to understand.
- Best for small projects.
- Suitable when requirements are fixed.

Disadvantages

- Inflexible.
- Difficult to make changes once development starts.

2. Agile Model

Agile is an **Incremental Model**.

The project is developed in small parts (iterations/sprints).

Features

- Incremental Development
- Flexibility
- Collaboration
- Continuous Feedback after deployment

Advantages

- Easy to handle changing requirements.
- Faster delivery.
- Continuous improvement through feedback.
- Better collaboration between client and team.

Disadvantages

- Requires continuous client involvement.
- Planning may change frequently.

- **Scrum**

Scrum is a framework used in Agile development.

It helps teams:

- Work in small iterations (Sprints)
- Deliver features quickly
- Collect feedback regularly
- Improve the product continuously

- **Which SDLC Model is Used by MNCs?**

Agile Model (Most Common)

Most **MNCs (Multinational Companies)** use the **Agile Model**, especially **Scrum**.

Why do MNCs prefer Agile?

- Requirements can change anytime.
- Faster delivery through **Sprints** (usually 2–4 weeks).
- Continuous client feedback.
- Better collaboration between developers, testers, and clients.
- Easy to fix bugs and add new features.

Examples of companies using Agile:

- Google
- Microsoft
- Amazon
- IBM
- Accenture
- Infosys
- TCS
- Wipro
- Capgemini
- Cognizant

Waterfall Model

Waterfall is still used in projects where requirements are fixed and changes are very rare.

Used in:

- Government projects
- Banking systems (some legacy projects)
- Healthcare systems
- Manufacturing projects
- Embedded systems

